MIL-P-43253B
10 January 1985
SUPERSEDING
MIL-P-43253A
2 September 1976

MILITARY SPECIFICATION

PUMP, LUBRICANT TRANSFER (HAND)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-operated lubricant transfer pump.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure Sensitive.
MIL-P-23377	- Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4930

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Threads Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipping and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 297 - Rubber Product - Chemical Analysis.
D 2000 - Rubber Products in Automotive Applications, Classification System for.
D 3951 - Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump shall be hand-held, hand-operated type, with a suction hose and a discharge hose (see figure 1).

3.2 First article. The contractor shall furnish one or more pumps as specified (see 6.2) for examination and testing (see 4.3) within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce pumps that comply with the requirements of this

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The pump shall be fabricated from compatible materials inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.4 Environmental.

3.4.1 Operating temperature. The pump shall operate in any ambient temperature from -30° F to +160° F when tested as specified herein.

3.4.2 Storage temperature. The pump shall not be damaged by storage in any ambient temperature from -65° F to +160° F.

3.5 Pump. The pump shall be reciprocating, hand held, hand operated, and self-priming, and shall weigh not more than 5 pounds. The pump shall discharge and intake simultaneously on every upward stroke after being primed, and shall transfer not less than 2 fluid ounces of lubricating oil per stroke.

3.5.1 Priming. The pump shall self-prime in not more than 4 strokes.

3.5.2 Operating force. The pump shall require a force of not more than 25 pounds (up or down stroke) to transfer 30wt (viscosity) oil which is at 70° F temperature.

3.6 Suction hose. The suction hose shall be polychloroprene rubber, conforming to ASTM D 2000, type M3BC610C12F17. Unless otherwise specified (see 6.2), the suction hose shall have an inside diameter of not less than .303 inch, a wall thickness of not less than .104 inch, and a length of not less than 12 inches. Suction hose shall be reinforced to prevent collapse when suction is applied.

3.7 Discharge hose. Unless otherwise specified (see 6.2), the discharge hose shall be identical to the suction hose, except that it shall not be less than 18 inches long. Discharge hose need not be reinforced to prevent collapse.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:00Z
Check the source to verify that this is the currentversion before use.

3.8 Threads. All threads shall be in accordance with FED-STD-H28.

3.9 Identification marking. The pumps shall be identified in accordance with MIL-STD-130.

3.10 Treatment and painting. The portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The portions of the pump normally not painted shall be protected from corrosion by a plating of zinc, nickel, or other electrodeposited coatings.

3.11 Workmanship. The pump shall be clean and free of any sharp edges or burrs and any defects which could affect its operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump shall be tested as specified in 4.5.2.1 through 4.5.2.3. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Acceptable quality level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 through 4.5.2.3. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pump shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Dimensions and weight not as specified.
105. Components missing or not as specified.
106. Treatment and painting not as specified.
107. Workmanship not as specified.
108. Discharge hose not as specified.
109. Threads not as specified.
110. Operating force not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.5.2 Tests.

4.5.2.1 Low temperature. The pump, with a sufficient amount of Number 10 weight oil, shall be equalized at -30° F, ±5° F. With the pump's intake no more than 1 inch below the oil level, self-priming shall be accomplished within 4 strokes of the pump. Following this, the pump's intake shall be immersed as necessary to discharge 10 full successive strokes into a graduated container. The pump shall not be damaged by storage in an ambient temperature of -65° F ±5° F. Nonconformance to 3.4 and 3.5 shall constitute failure of this test.

4.5.2.2 High temperature. The pump, with a sufficient amount of Number 30 weight oil, shall be equalized at +160° F, ±5° F. With the pump's intake no more than 1 inch below the oil level, self-priming shall be accomplished within 4 strokes of the pump. Following this, the pump's intake shall be immersed as necessary to discharge 10 full successive strokes into a graduated container. The pump shall not be damaged by storage in an ambient temperature of 160° F ±5° F. Nonconformance to 3.4 and 3.5 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:00Z
Check the source to verify that this is the currenversion before use.

4.5.2.3 Rubber compound. Rubber compound shall be tested for polychloroprene in accordance with ASTM D 297. Nonconformance to 3.6 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

111.	Preservation not as specified for level A.
112.	Box not as specified for level A packing.
113.	Gross weight of shipping container exceeds limitations specified for levels A and B.
114.	Box not as specified for level B.
115.	Shipping containers not closed and strapped as specified for levels A and B.
116.	Marking illegible, incorrect, or missing for all degrees of packaging.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The pump shall be disassembled according to the manufacturer's recommendations to reduce cube to the minimum. Preservation shall be in accordance with MIL-P-116, method III, using a snug fitting box conforming to PPP-B-636, class weather-resistent, grade W6c, style optional. The box shall be closed as specified for method V of the appendix to the box specification.

5.1.2 Commercial. The pump shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Pumps, preserved as specified in 5.1, shall be packed together in snug-fitting boxes conforming to PPP-B-601, overseas type, style optional. Quantities packed shall be such that the gross weight of the box does not exceed 250 pounds. Box closure and reinforcement shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781 and shall be finish B unless otherwise specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Pumps, preserved as specified in 5.1, shall be packed in snug-fitting boxes conforming to PPP-B-636, class weather resistant, grades V3c, V11c or V13c, style optional, in quantities not to exceed the size or weight limitations of the box. Boxes shall be closed and reinforced in accordance with the appendix to the box specification. Single pumps being shipped, preserved as specified in 5.1.1, shall require no additional packing.

5.2.3 Commercial. Pumps, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Levels A and B. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on each shipping container.

6. NOTES

6.1 Intended use. The pump is intended for draining crankcase oil from engines, for use as an inflation pump, and for transferring various types of fuel and oil.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article pump, and number of pumps required (see 3.2).
c. Suction hose required if other than as specified (see 3.6).
d. Discharge hose required if other than as specified (see 3.7).
e. Degree of preservation and packaging required (see 5.1 and 5.2).
f. When strapping shall be other than as specified (see 5.2.1).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents' first articles.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.5 Information figures. Figure 1 shows a type of pump which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another pump which conforms to this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

6.6 Classification change. Type II pumps have been discontinued. The type II pumps are now covered by XX-D-370.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - MD
DLA - CS

Preparing activity:
Army - ME

Project 4930-0315

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

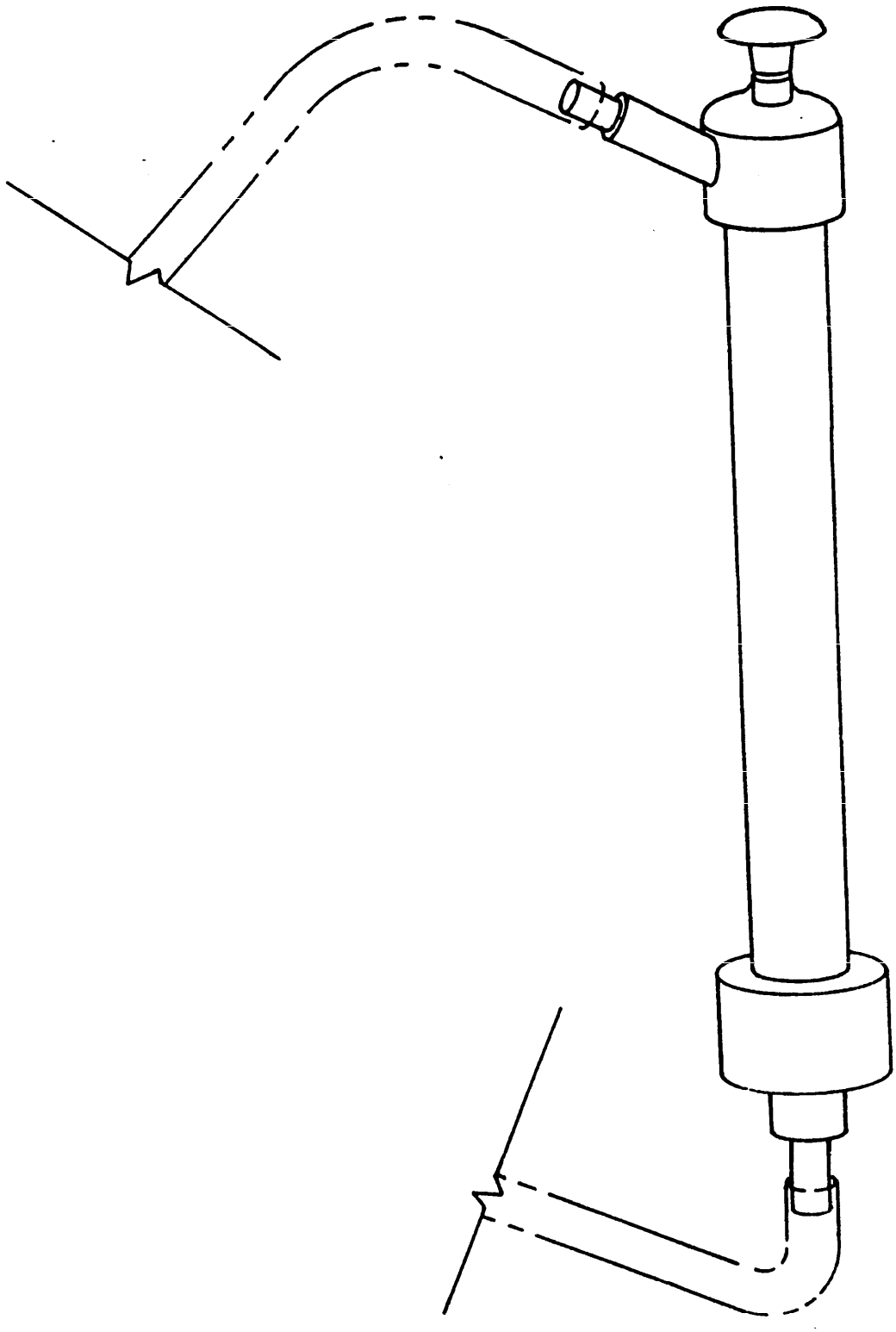


FIGURE 1. Pump, lubricant transfer (hand).

X-3160

U.S. GOVERNMENT PRINTING OFFICE: 1985 - 505-038/20704

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T21:00Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir R & D Center
ATTN: STRBE-DS
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-43253B	Pump, Lubricant Transfer (Hand)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:00Z
Check the source to verify that this is the current version before use.